Input Ontology (IO)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *MATHDIAL* | ***Context:*** *Indian Metropolitan Professional Teachers (Grades 1–8)*

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The user explicitly confirmed that curriculum-agnostic math errors and the benchmark's eight pedagogical categories are necessary and sufficient for the deployment task [A1, A3], substantially reducing ontological misalignment risk. The taxonomy is theoretically grounded in Reiser's (2004) scaffolding framework distinguishing structure (Focus) from problematize (Probing) moves [Q36, Q37], with Telling and Generic as additional categories [Q40, Q41]. The dataset confirms transparent application of the four-move taxonomy across all sampled dialogues, providing structured pedagogical vocabulary [DATASET-D2, DATASET-D3]. Confusion type coverage spans six algebra-misconception categories [DATASET-D7, DATASET-D8, DATASET-D10]. Minor concerns remain: the taxonomy's structural privileging of scaffolding over directive correction [Q154, Q155] embeds a constructivist orientation that, while not omitting categories the user requires, frames the relationships among them in a way that may not match Indian metro teaching priorities. The benchmark also explicitly omits meta-cognitive support and rapport-building [Q112], though the user did not flag these as required.

ID	Evidence
Q36	“We base the first two on the work of Reiser (2004), who suggest that scaffolding strategies can be split into two main categories: structure and problematize.” (p.5)
Q37	“These form the basis for the Focus and Probing moves employed in our study.” (p.5)
Q40	“This is called Telling.” (p.5)
Q41	“Finally, turns that just serve as conversational elements and have limited pedagogical value are classed as Generic.” (p.5)
Q112	“Inspired by previous work in scaffolding, we acknowledge our focus is on a subset of common teaching moves. However, this does not cover all the goals of human tutors, such as meta-cognitive support or building rapport with a student.” (p.10)
Q154	“Most importantly, scaffolding questions that are productive for long-term learning are Focus and Probing.” (p.17)
Q155	“On the other hand, Telling represents giving out the partial or full answer to the student and should be mostly used when a student is stuck.” (p.17)
D2	Hi Rishi. Can you tell me how many hours it would take Apple to run 24 miles?" ... "Good. How many hours would it take Mac?" ... "How much longer does it take Apple?" ... "How much longer is that in minutes?" ... "So how much faster is Mac then Apple in minutes? — Six consecutive probing questions guiding student to answer; entirely Socratic, no direct correction
D3	Carmela has 600*2 + 50 = 1250" ... "It is 1250. Not 2450" ... "600+950+1250 = 2800 — Teacher resorts to Telling only after multiple Focus moves fail; Telling framed as reluctant fallback
D7	Student: Sure. In the first half of the year, Nicki ran for 26 weeks. Since I wanted to calculate how many miles she ran per week, I divided the total number of weeks by 2 to get the average number of miles she ran per week. — Simulated student produces unprompted multi-clause explanation of own reasoning process
D8	Student: I divided the total number of desserts (126) by the number of people in the family (7) to find out how many desserts each person would get (18). Then, I divided 18 by 3, since there are 3 types of desserts, to find out how many of each dessert each person would get (6). — Unprompted full explanation of two-step reasoning from student
D10	Student profile: Winnie is a 7th grade student. She struggle most with understanding what the problem is asking them to do. — US-style student name and brief English-language confusion profile; no Indian cultural markers

Strengths

- Eight pedagogical categories user-confirmed as necessary and sufficient for the deployment task [A3]
- Theoretically anchored in scaffolding learning sciences (Reiser 2004) [Q36, Q37]

- Fine-grained move labels systematically applied to every teacher utterance, enabling multi-dimensional pedagogical evaluation [DATASET-D2, DATASET-D3]
- Broad coverage of student confusion types relevant to Grades 1–8 math reasoning [DATASET-D7, DATASET-D8, DATASET-D10]

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Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: User confirmed curriculum-agnostic math errors and the eight pedagogical categories suffice for the Indian metro deployment [A1, A3]. Required test categories include Focus, Probing, Telling, Generic moves and conceptual/procedural error confusions, all present in the benchmark.

IO-2: No regionally-relevant categories are identified by the user as missing. The benchmark omits meta-cognitive support, rapport-building [Q112], and visual instructional practices [Q113], but the user did not flag these as required for the deployment.

IO-3: No categories are identified as irrelevant to the regional context. All four move types (Focus, Probing, Telling, Generic) appear in real classroom practice and Indian metro teachers can evaluate them, though the framing of Telling as a last resort [Q155] may be at odds with Indian pedagogical norms (handled under OO).

IO-4: No major content-validity gaps identified at the ontology level given user confirmation of sufficiency [A3]. Residual concern is that the taxonomy's hierarchical valuation (scaffolding productive, Telling last-resort) [Q154, Q155] is itself a value-laden ontological choice.

ID	Evidence
Q42	“Table 2 lists finer-grained intents for each of these four categories along with a set of accompanying examples.” (p.5)
Q112	“Inspired by previous work in scaffolding, we acknowledge our focus is on a subset of common teaching moves. However, this does not cover all the goals of human tutors, such as meta-cognitive support or building rapport with a student.” (p.10)

Q113	“Moreover, text tutoring limits teachers' use of additional instructional practices such as drawings.” (p.10)
Q154	“Most importantly, scaffolding questions that are productive for long-term learning are Focus and Probing.” (p.17)
Q155	“On the other hand, Telling represents giving out the partial or full answer to the student and should be mostly used when a student is stuck.” (p.17)

Information Gaps

- Whether the user's acceptance of the eight categories was made with full awareness of how Telling is structurally devalued in the framework

Requires Expert Verification

- Whether Indian metro teachers, after reviewing benchmark dialogues, would still affirm the eight categories as sufficient or would flag missing categories (e.g., explicit demonstration, drill, exam-strategy guidance)

Remediation

Gap 1: Although user accepted the eight categories, the taxonomy embeds a hierarchical valuation (scaffolding 'productive', Telling 'last resort') that may not survive scrutiny once Indian teachers see the benchmark dialogues.

Recommendation: Conduct a confirmatory taxonomy-elicitation session with 5–10 Indian metro teachers after they review 20+ benchmark dialogues, asking whether they would add categories (e.g., explicit demonstration, drill, exam-strategy guidance) or merge existing ones. Update taxonomy if needed before deployment.